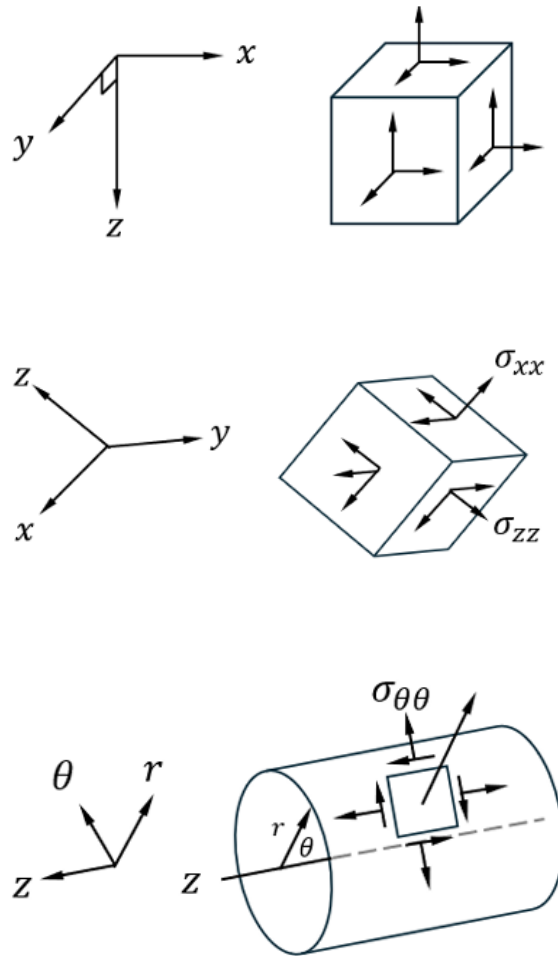


Question 1

Label the stress components acting on a differential element in two dimensions.



Question 2

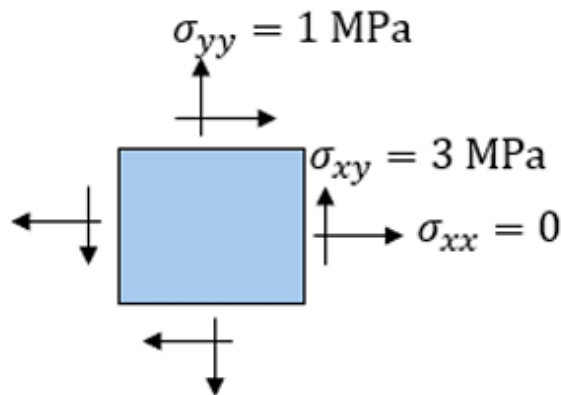
For a point P in a two-dimensional stress field, explain the meanings of σ_{xx} , σ_{yy} , σ_{xy} , σ_n , and σ_{ns} , where $\mathbf{n}$ is the unit normal to a plane.

Question 3

- What is meant by the stress state of a point?
- List the components required to define it in two dimensions.

Question 4

At a certain point in a two-dimensional stress field, the normal and shear stresses are as shown in the figure below.



- Determine the values of the maximum and minimum normal stresses at the point and show the directions of the principal stresses on a sketch.
- Determine the value of the maximum shear stress and its direction at the point.

Question 5

Write down the equations of motion in the following format.

- Index notation.
- Unabridged notation.

Question 6

At a certain point in a 2-D stress field,

$$\sigma_{xx} = 0, \sigma_{xy} = -\sqrt{3}/2, \sigma_{yy} = \sqrt{3}$$

determine

- the values of the maximum and minimum normal stresses at the point and
- show the directions of the principal stresses on a sketch.

Question 7

Apply Newton's 2nd law to a differential material cubic to derive the following x-direction stress equation of motion

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{yx}}{\partial y} + \frac{\partial \sigma_{zx}}{\partial z} = \rho a_x$$

and state any assumption made.

